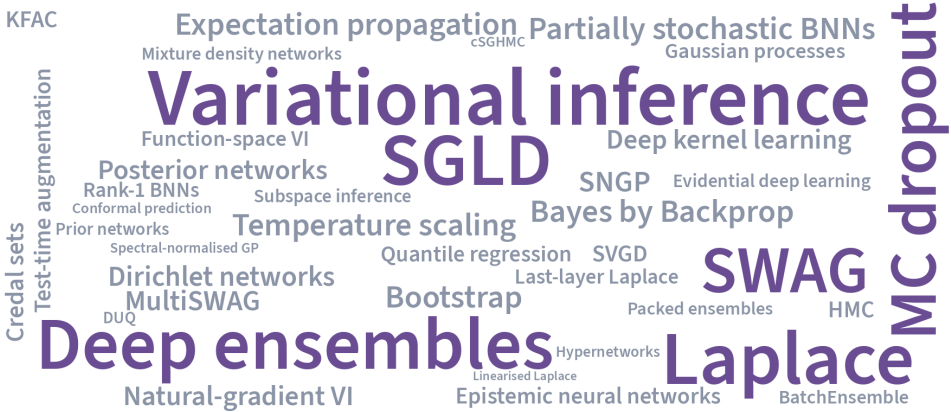
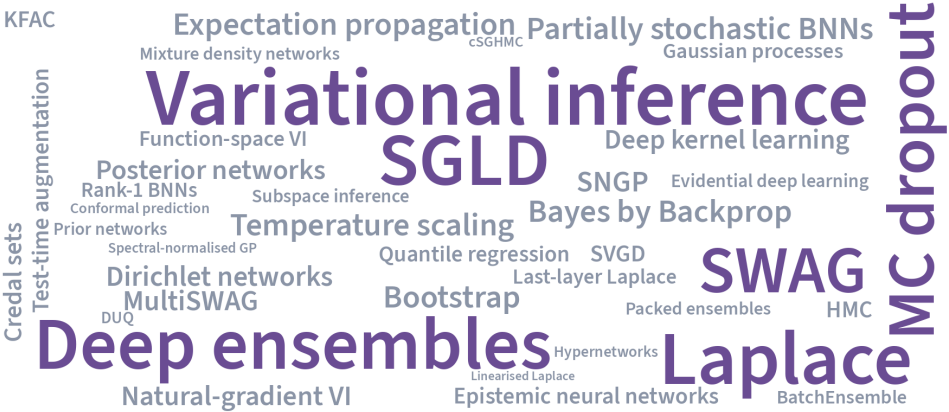
ACCELERATE PROGRAMME FOR SCIENTIFIC DISCOVERY

Uncertainty in AI

Why this workshop?



Why this workshop?



Today: their context, what they do, and where they fail

Why this workshop?

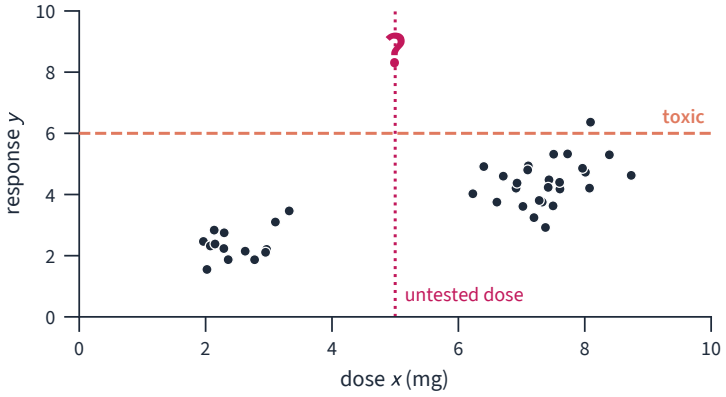
Uncertainty methods are one piece of a larger puzzle.

Why this workshop?

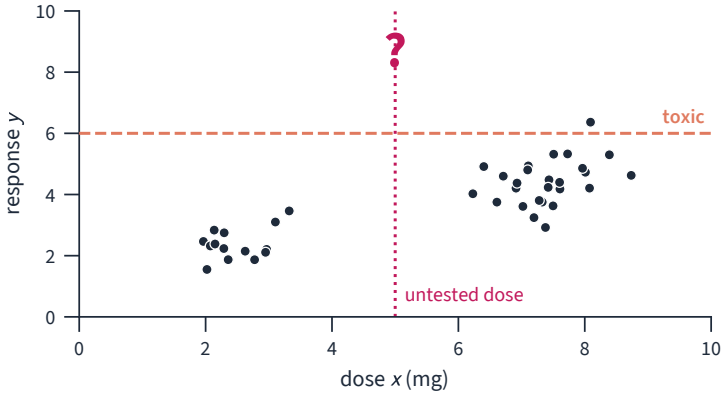
Uncertainty methods are one piece of a larger puzzle.

To understand their limitations,
we need the whole picture.

Should we approve this dose?



Should we approve this dose?



approve reject run more trials

Uncertainty in AI

Uncertainty in AI

What is AI?

“The sciences do not try to explain, they hardly even try to interpret, they mainly make models. By a model is meant a mathematical construct which, with the addition of certain verbal interpretations, describes observed phenomena.”

— John von Neumann

J. von Neumann (1955). Method in the physical sciences. In L. Leary (ed.), *The Unity of Knowledge*.

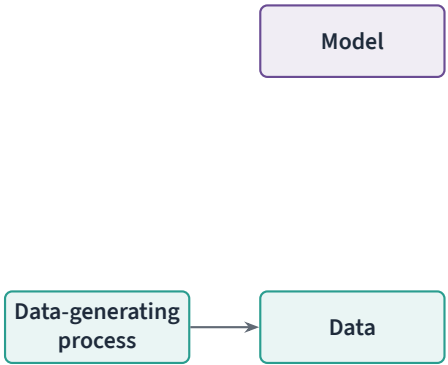
AI in one picture

In this workshop: models of observed phenomena, used to predict and decide



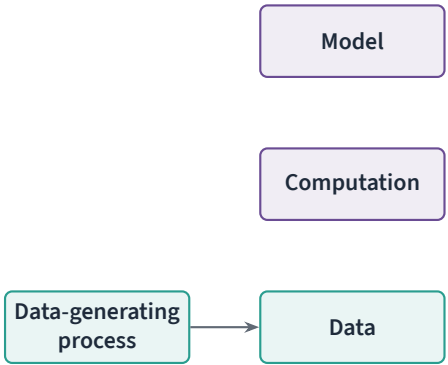
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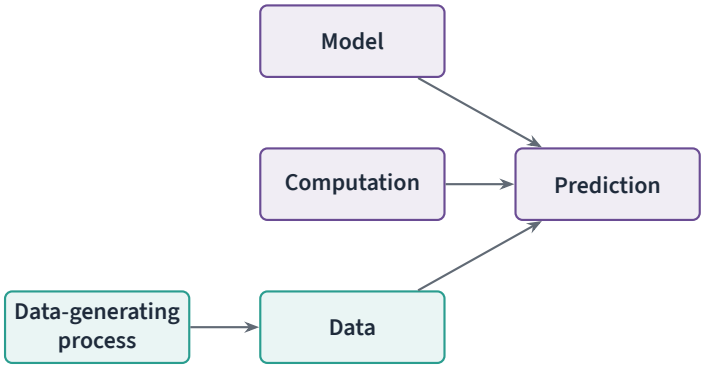
AI in one picture

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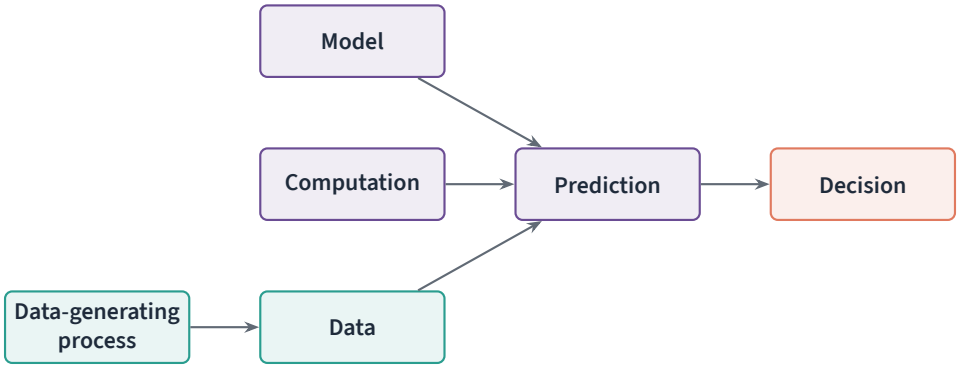
AI in one picture

In this workshop: models of observed phenomena, used to predict and decide



AI in one picture

In this workshop: models of observed phenomena, used to predict and decide



Uncertainty in AI

Knowing what we don't know

The Oracle at Delphi declared that no one was wiser than Socrates.

Plato, *Apology* 21a–d, translated by B. Jowett (1871).

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"He knows nothing, and thinks that he knows; I neither know nor think that I know."

— Socrates, in Plato's *Apology*

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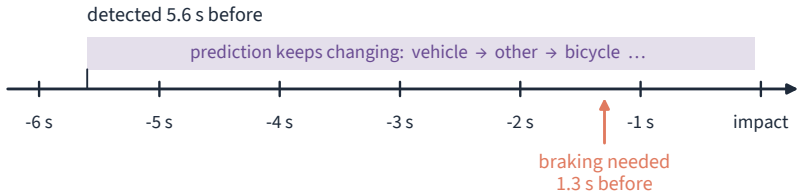
— Socrates, in Plato's *Apology*

We want models that know what they don't know

Plato, *Apology* 21a–d, translated by B. Jowett (1871).

When a model doesn't know what it doesn't know

A self-driving test car hits and kills a woman walking her bicycle across the road

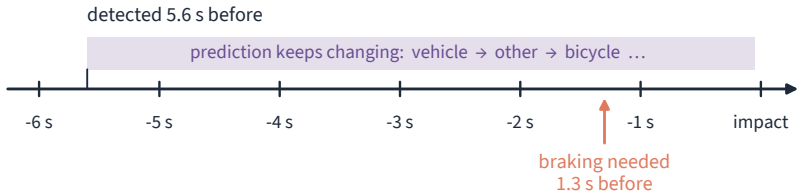


its software had no notion of a pedestrian outside a crosswalk

Tempe, Arizona, 2018. NTSB (2019), HAR-19/03; other failures contributed too.

When a model doesn't know what it doesn't know

A self-driving test car hits and kills a woman walking her bicycle across the road



its software had no notion of a pedestrian outside a crosswalk

Uncertain and unfamiliar should have meant: slow down

Tempe, Arizona, 2018. NTSB (2019), HAR-19/03; other failures contributed too.

Uncertainty changes decisions

Should you take an umbrella?

chance of rain: 5% → leave it at home

Uncertainty changes decisions

Should you take an umbrella?

chance of rain: 5%	→	leave it at home
chance of rain: 80%	→	take it

Uncertainty changes decisions

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The probability alone changes the decision

Probability: joint, marginal, conditional

the joint $p(A, B)$ A and B together; it all adds up to 1

joint $p(A, B)$

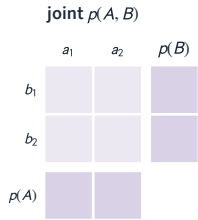
	a_1	a_2
b_1		
b_2		

Probability: joint, marginal, conditional

the joint $p(A, B)$ A and B together; it all adds up to 1

the marginal $p(A)$ A, whatever B is

sum rule: $p(A) = \sum_B p(A, B), \quad p(B) = \sum_A p(A, B)$



Probability: joint, marginal, conditional

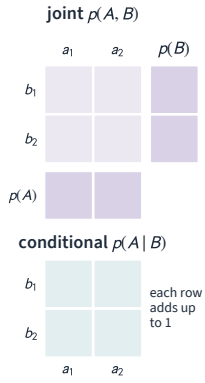
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product rule: $p(A, B) = p(A \mid B)\, p(B)$, so $p(A \mid B) = \dfrac{p(A, B)}{p(B)}$



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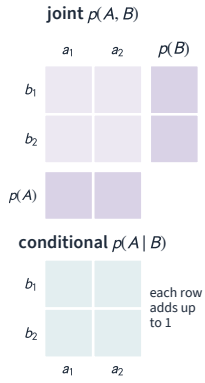
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Bayes' rule: write the product rule both ways,

$p(A \mid B)\, p(B) = p(B \mid A)\, p(A)$, so $p(B \mid A) = \dfrac{p(A \mid B)\, p(B)}{p(A)}$



Probability in practice: a case for Sherlock Holmes

A wealthy widow is found dead in her study. That night, only the butler and the gardener were on the estate. The butler had argued with her that morning.

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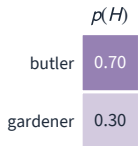
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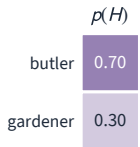
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Before looking for any clue: who do you suspect?

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Once Holmes checks for petals, how should he update his initial belief?

Bayes' rule: updating our beliefs

$$p(H \mid E) = \frac{p(E \mid H)\, p(H)}{p(E)}$$

Bayes' rule: updating our beliefs

$$p(H \mid E) = \frac{p(E \mid H)\, p(H)}{p(E)}$$

$p(H)$	**the prior**: our initial belief, before the clue
$p(E \mid H)$	how likely the clue is, for each culprit
$p(E)$	how likely the clue is overall
$p(H \mid E)$	**the posterior**: our updated belief, after the clue

Bayes' rule: updating our beliefs

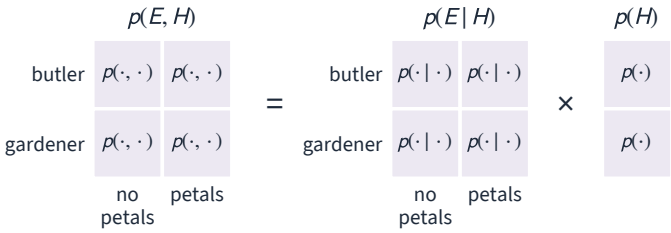
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$p(H \mid E)$	**the posterior**: our updated belief, after the clue

Step 1: the joint	$p(E, H) = p(E \mid H)\, p(H)$	the product rule
Step 2: the marginal	$p(E) = \sum_H p(E, H)$	the sum rule
Step 3: the posterior	$p(H \mid E) = p(E, H) \,/\, p(E)$	divide

Building the posterior: the joint, via the product rule

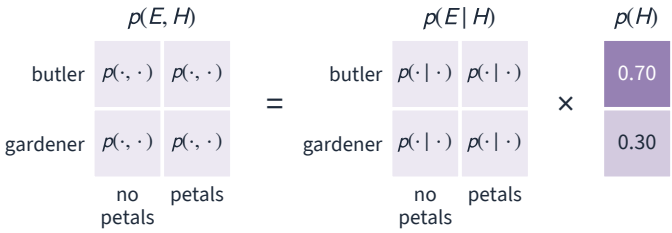
$$p(E, H) = p(E \mid H)\, p(H)$$



the joint is hard to write down directly; the product rule splits it into simpler pieces

Building the posterior: the joint, via the product rule

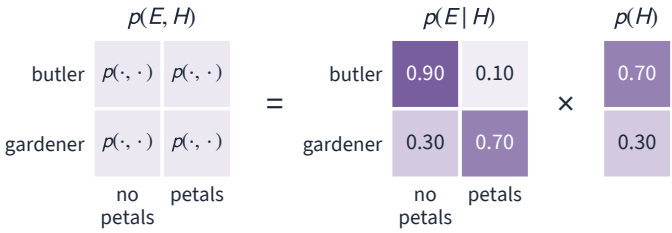
$$p(E, H) = p(E \mid H)\, p(H)$$



$p(H)$: the initial belief, before any clue

Building the posterior: the joint, via the product rule

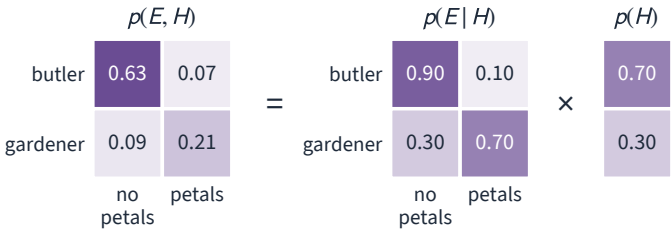
$$p(E, H) = p(E \mid H)\, p(H)$$



$p(E \mid H)$: how likely petals are, for each culprit

Building the posterior: the joint, via the product rule

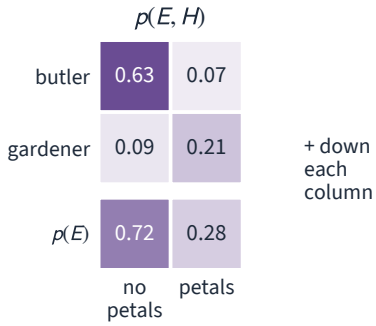
$$p(E, H) = p(E \mid H)\, p(H)$$



multiply, row by row: the joint

Building the posterior: the marginal, via the sum rule

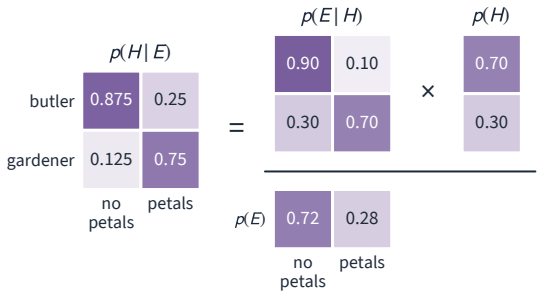
$$p(E) = \sum_H p(E, H)$$



add up each column: how likely each clue is overall

Bayes' rule: updating our beliefs

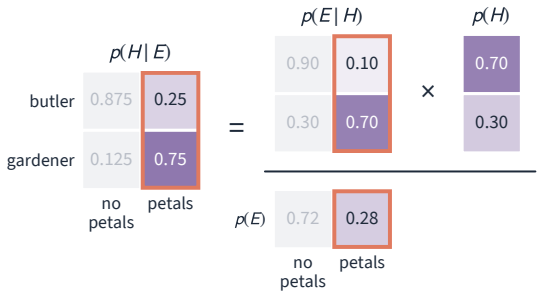
$$p(H \mid E) = \frac{p(E \mid H)\, p(H)}{p(E)}$$



before looking: what each clue would mean

Bayes' rule: updating our beliefs

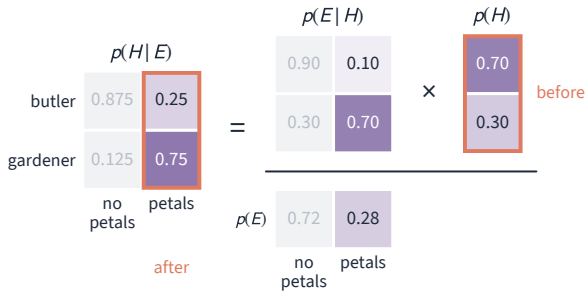
$$p(H \mid E) = \frac{p(E \mid H)\, p(H)}{p(E)}$$



Holmes finds petals: only the petals column matters

Bayes' rule: updating our beliefs

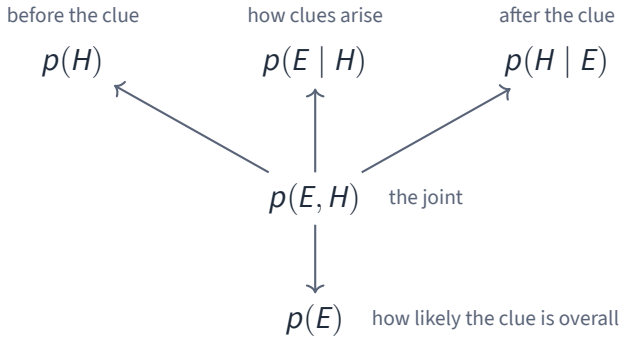
$$p(H \mid E) = \frac{p(E \mid H)\, p(H)}{p(E)}$$



from the initial belief to the updated belief

Our belief is updated: from 70% butler to 75% gardener

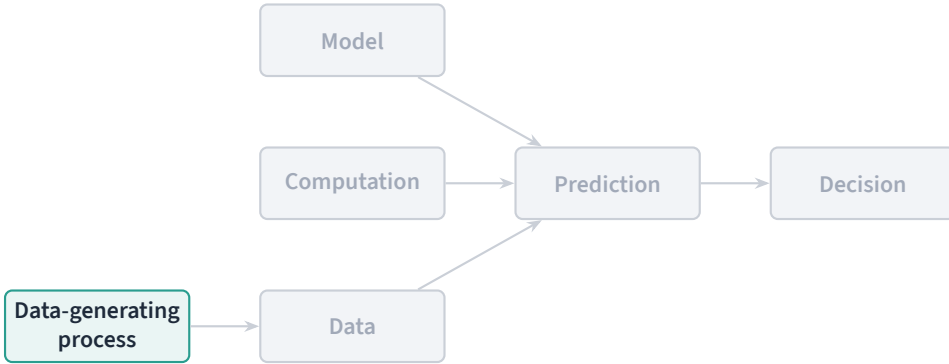
What we just did



Everything follows from the joint, with the sum and product rules

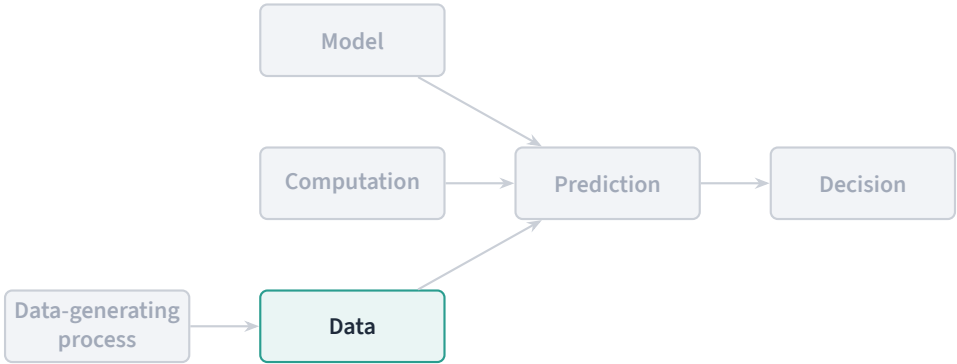
Uncertainty in AI

Where does uncertainty come from?



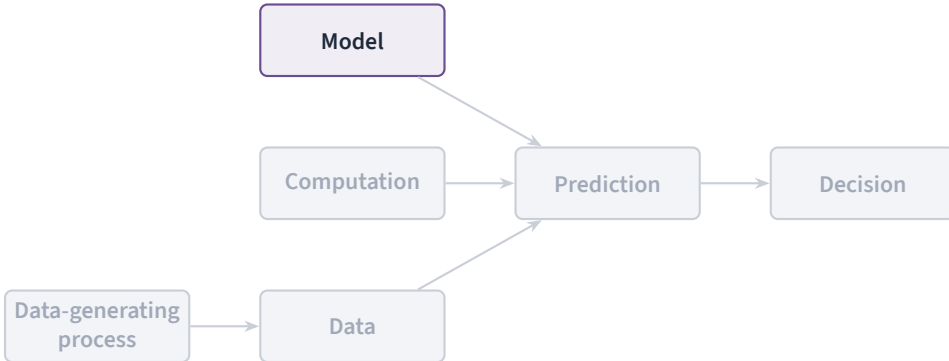
The phenomenon itself may be random

Where does uncertainty come from?



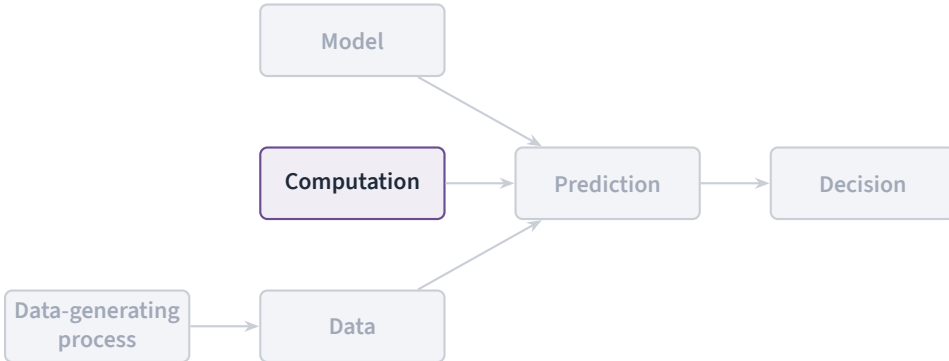
What, how, and how much we measure

Where does uncertainty come from?



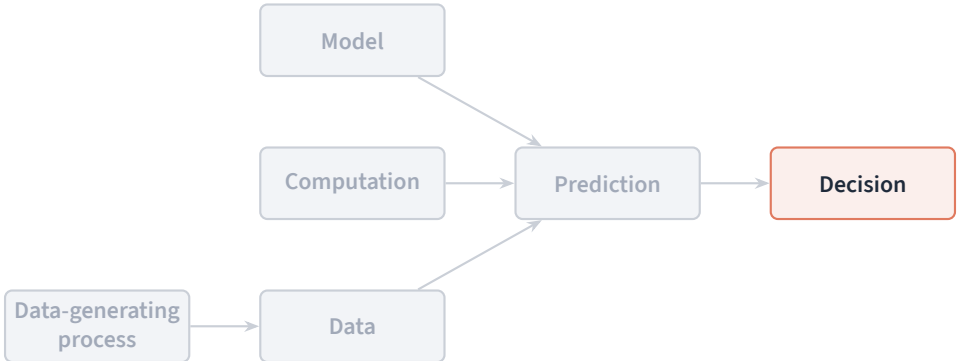
What to model, what kind of model, and which settings of it

Where does uncertainty come from?



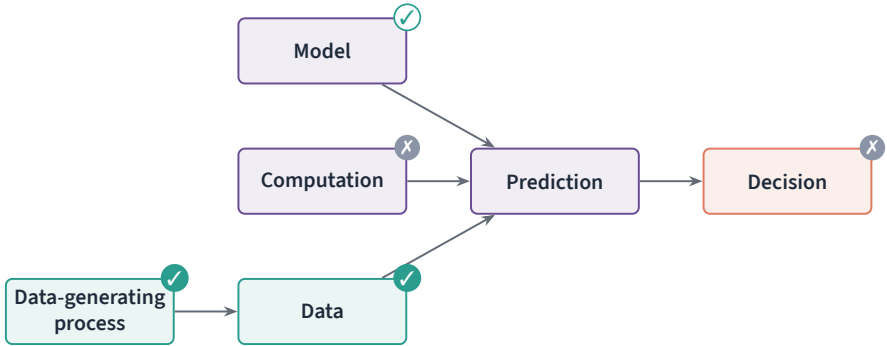
Shortcuts we take to get answers in reasonable time

Where does uncertainty come from?



What do we truly care about?

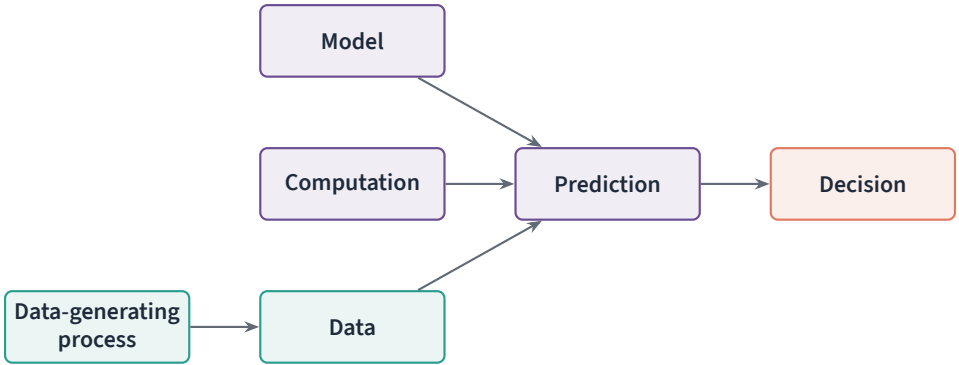
This workshop



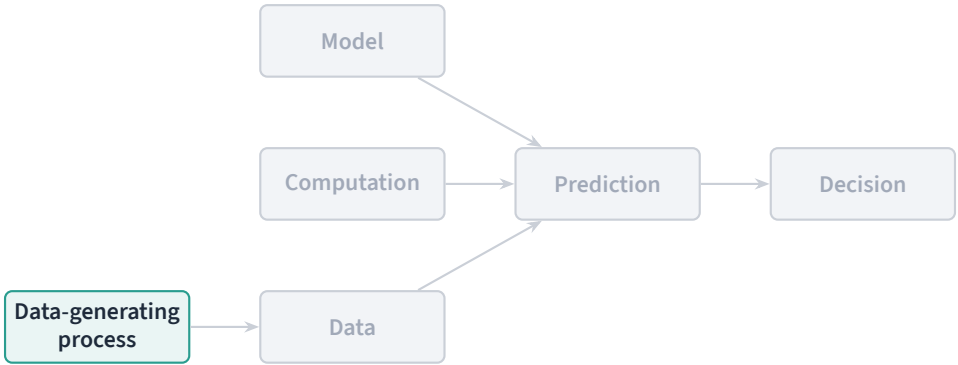
We capture randomness in the process and limited data, which shows up as uncertainty about the model's settings

✓ captured ✓ partly ✗ not captured

Next: the data-generating process

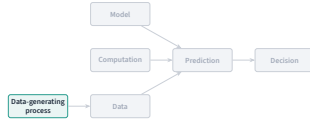


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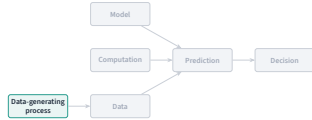
The phenomenon itself may be random

Many phenomena can be described by a joint



$$p_{\text{true}}(x, y) = p_{\text{true}}(y \mid x)\, p_{\text{true}}(x)$$

Many phenomena can be described by a joint



$$p_{\text{true}}(x, y) = p_{\text{true}}(y \mid x)\, p_{\text{true}}(x)$$

x	the dose a patient receives
y	the patient's response

Many phenomena can be described by a joint

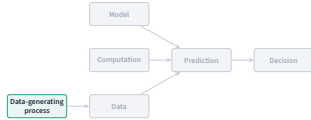


$$p_{\text{true}}(x, y) = p_{\text{true}}(y \mid x)\, p_{\text{true}}(x)$$

x	the dose a patient receives
y	the patient's response
$p_{\text{true}}(x)$	which doses are given
$p_{\text{true}}(y \mid x)$	how patients respond to a given dose

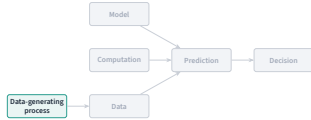
The joint describes both which doses are given and how patients respond

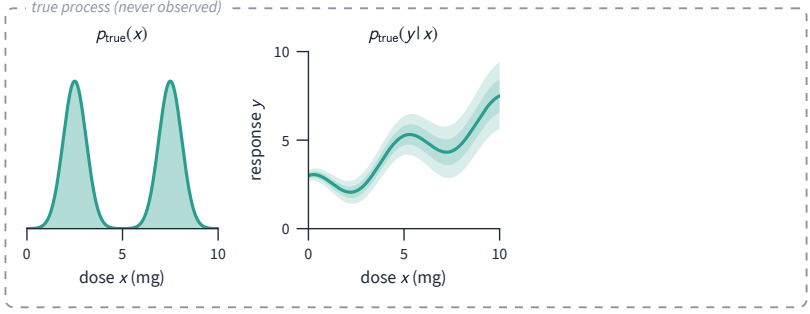
Thought experiment 1



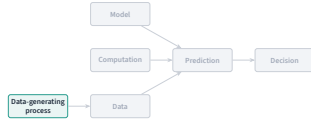


Thought experiment 1

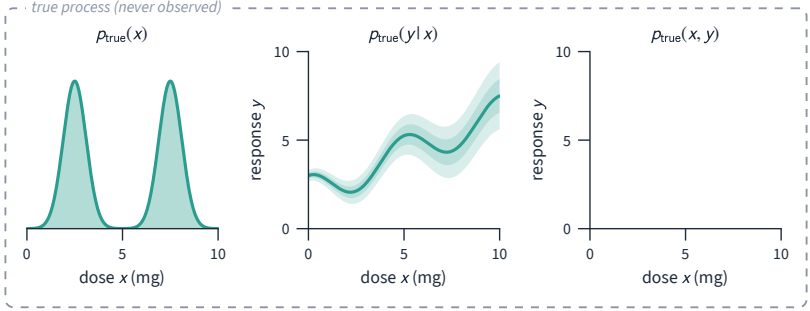




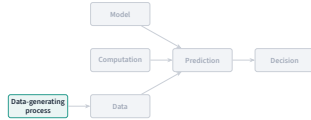
Thought experiment 1



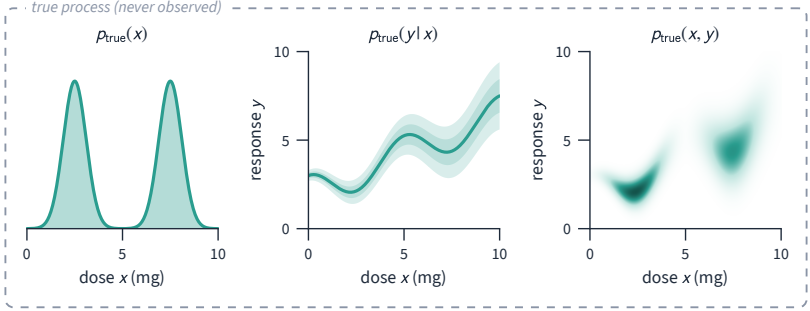
PREDICT What does the joint $p_{\text{true}}(x, y)$ look like?



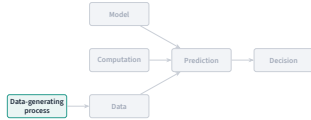
Thought experiment 1

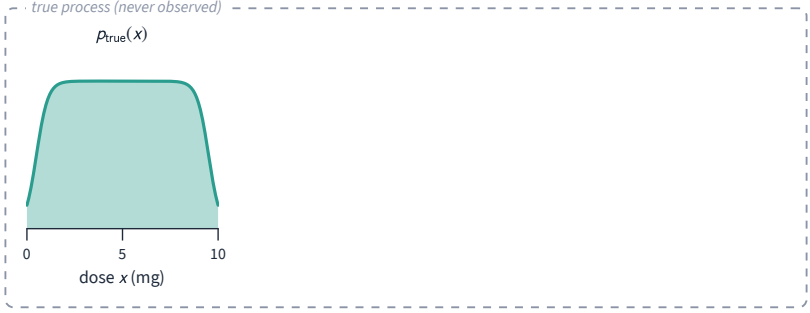


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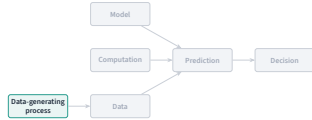


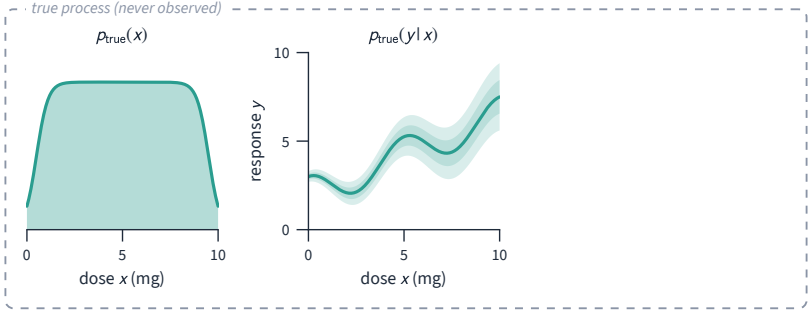
Thought experiment 2



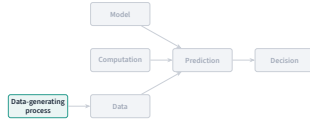


Thought experiment 2

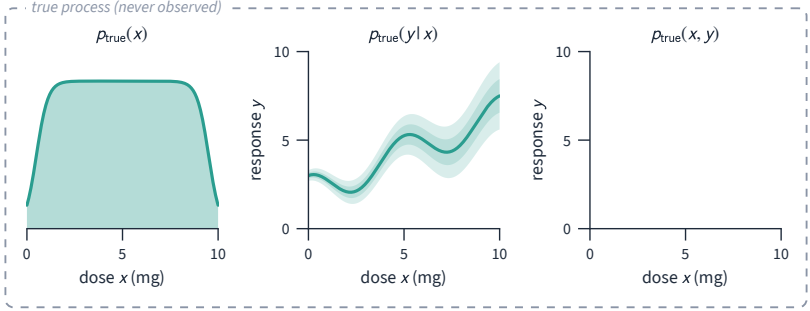




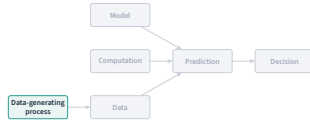
Thought experiment 2



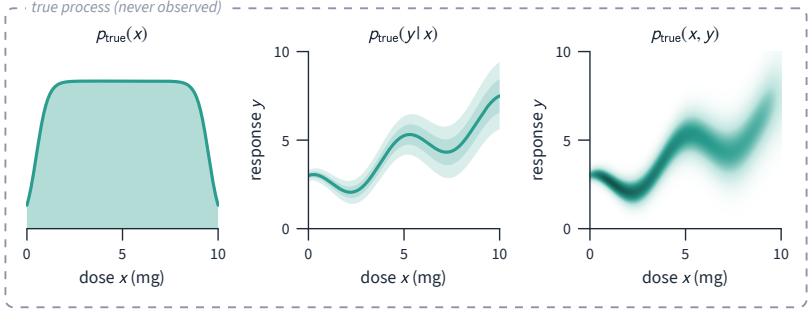
PREDICT What does the joint $p_{\text{true}}(x, y)$ look like?



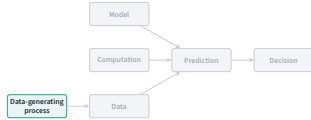
Thought experiment 2

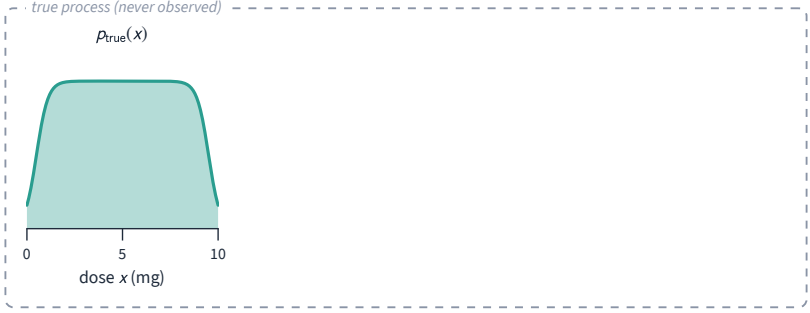


PREDICT What does the joint $p_{\text{true}}(x, y)$ look like?

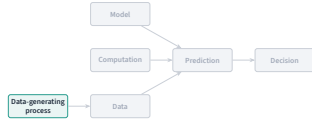


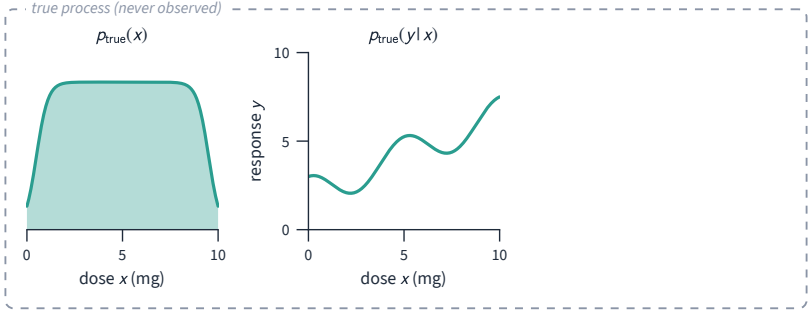
Thought experiment 3



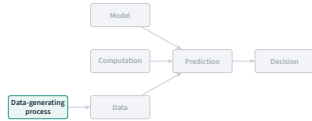


Thought experiment 3

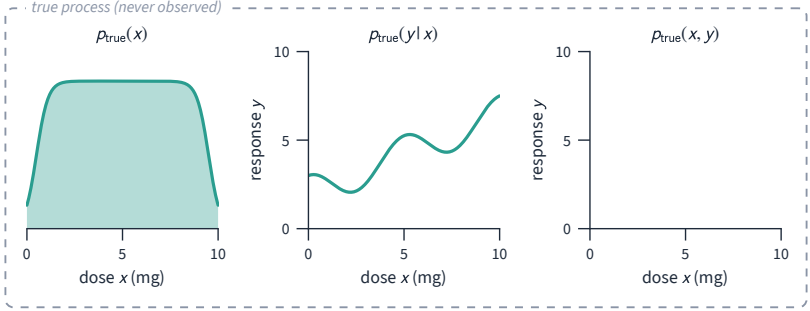




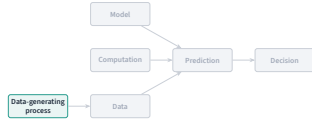
Thought experiment 3



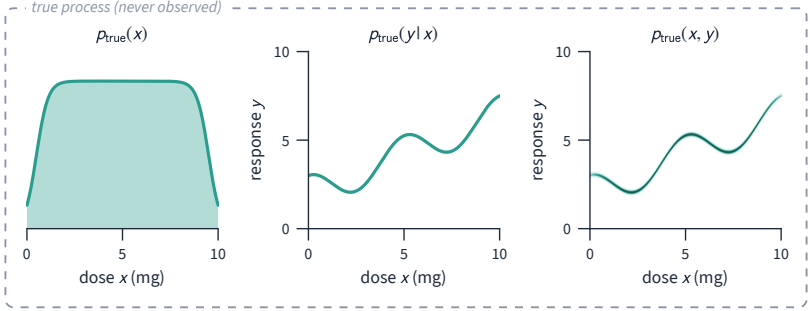
PREDICT What does the joint $p_{\text{true}}(x, y)$ look like?



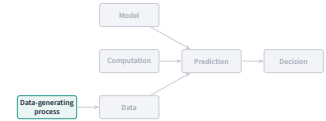
Thought experiment 3



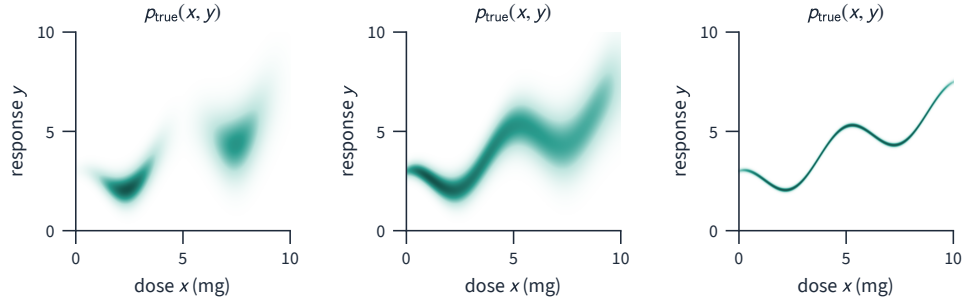
PREDICT What does the joint $p_{\text{true}}(x, y)$ look like?



Three processes, one rule



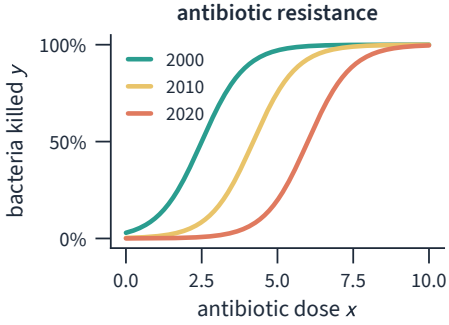
true process (never observed)

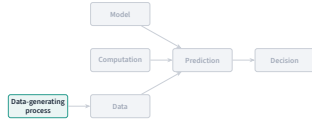


$$p_{\text{true}}(x, y) = p_{\text{true}}(y \mid x)\, p_{\text{true}}(x)$$

$p(x)$: which doses matter **$p(y \mid x)$: how variable the responses are**

When the world changes



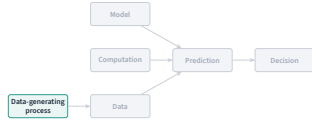


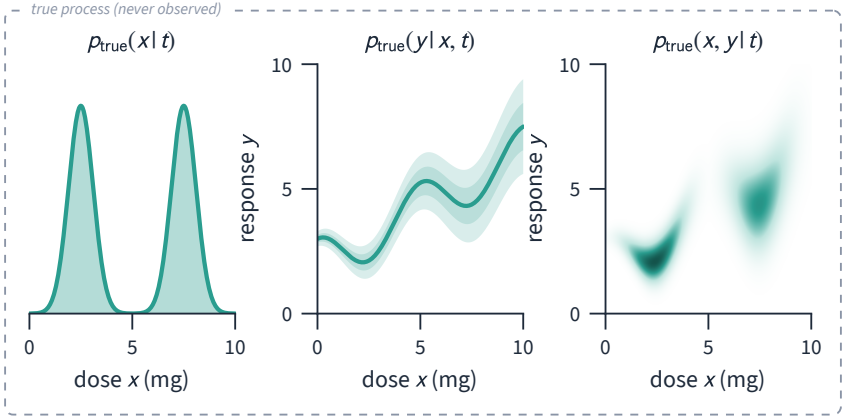
$$p_{\text{true}}(x, y \mid t)$$

the process can change with time t

antibiotics: the same dose kills fewer bacteria every decade, as resistance evolves

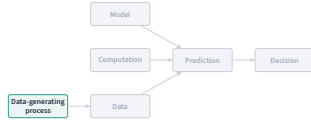
When the world changes: covariate shift

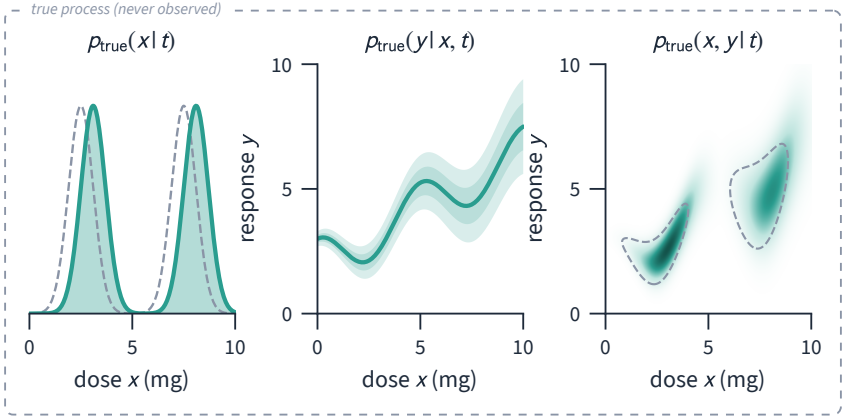




$t = 0$

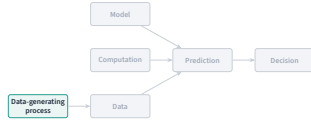
When the world changes: covariate shift

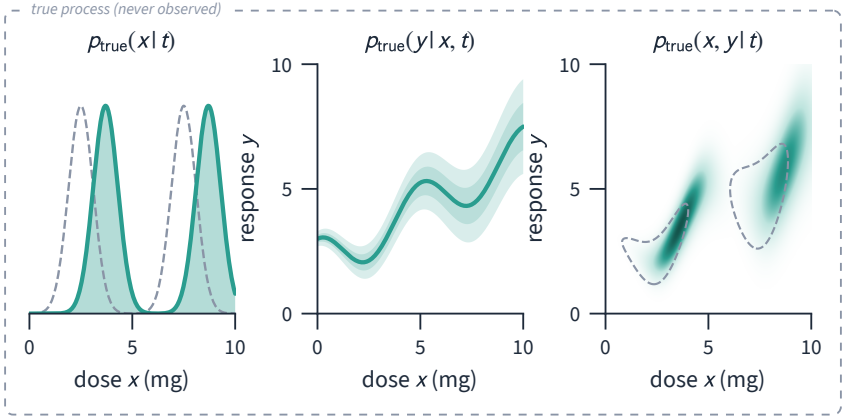




$t = 1$ **Doses drift: $p(x \mid t)$ moves, $p(y \mid x, t)$ stays**

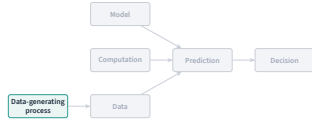
When the world changes: covariate shift

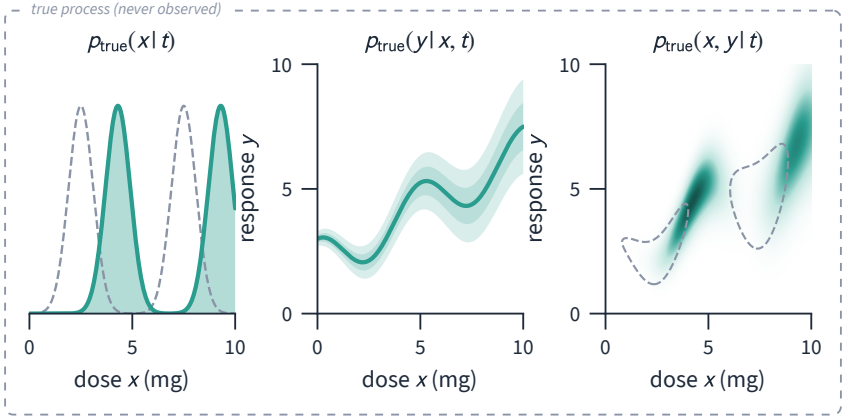




$t = 2$ **Doses drift: $p(x \mid t)$ moves, $p(y \mid x, t)$ stays**

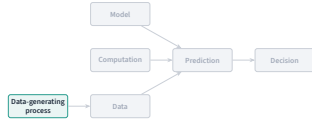
When the world changes: covariate shift

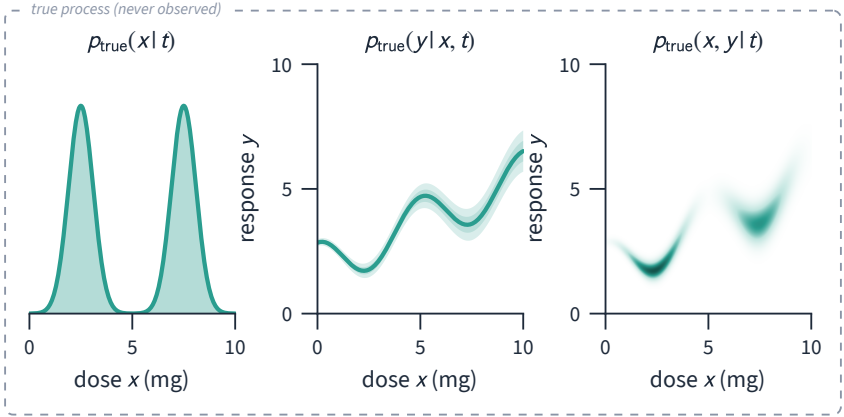




$t = 3$ **Doses drift: $p(x \mid t)$ moves, $p(y \mid x, t)$ stays**

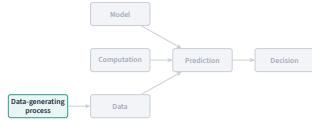
When the world changes: concept shift

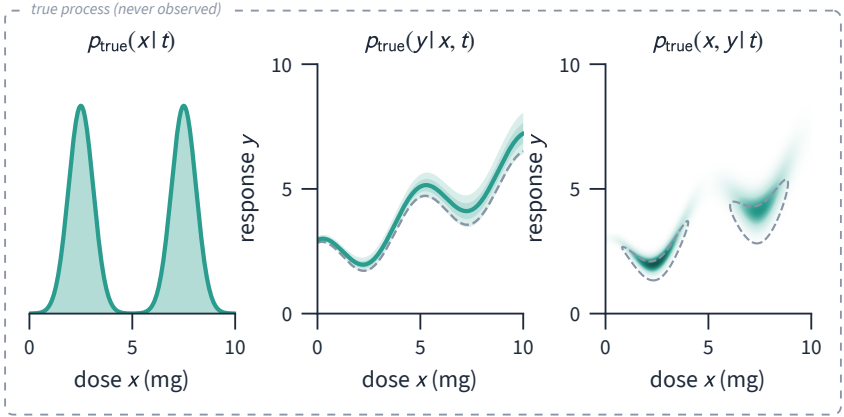




$t = 0$

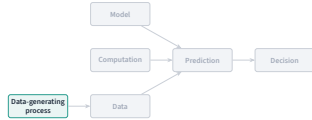
When the world changes: concept shift

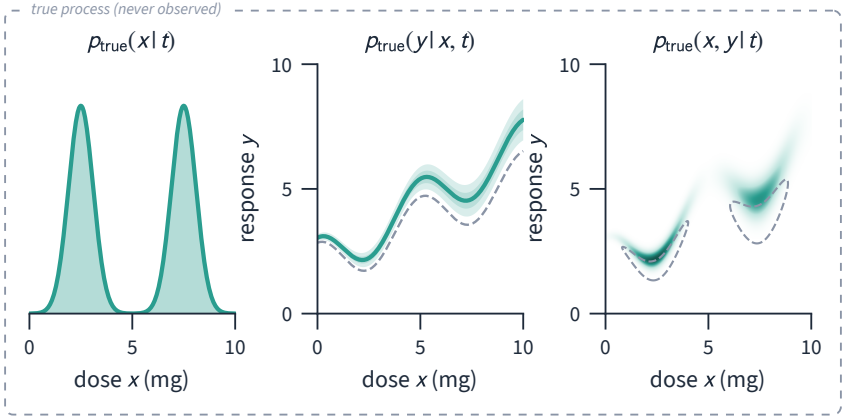




$t = 1$ **Responses drift:** $p(x \mid t)$ **stays,** $p(y \mid x, t)$ **changes**

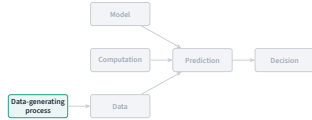
When the world changes: concept shift

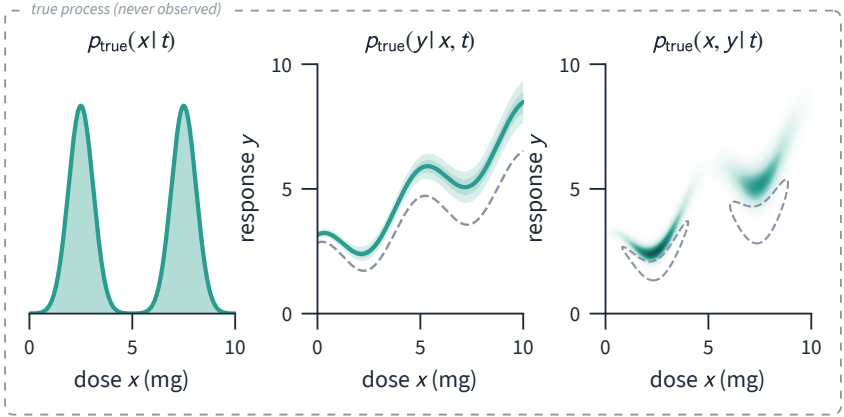




$t = 2$ **Responses drift:** $p(x \mid t)$ **stays,** $p(y \mid x, t)$ **changes**

When the world changes: concept shift





$t = 3$ **Responses drift:** $p(x \mid t)$ **stays,** $p(y \mid x, t)$ **changes**

Discussion: your own data

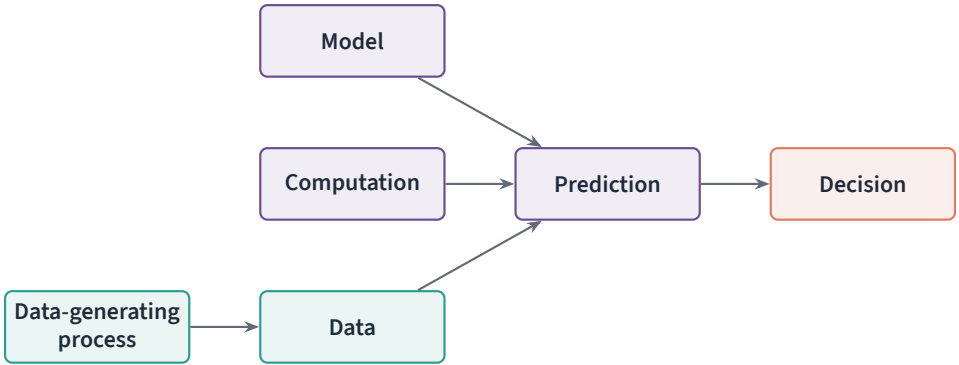


What is the joint distribution $p_{\text{true}}(x, y)$ in your work?

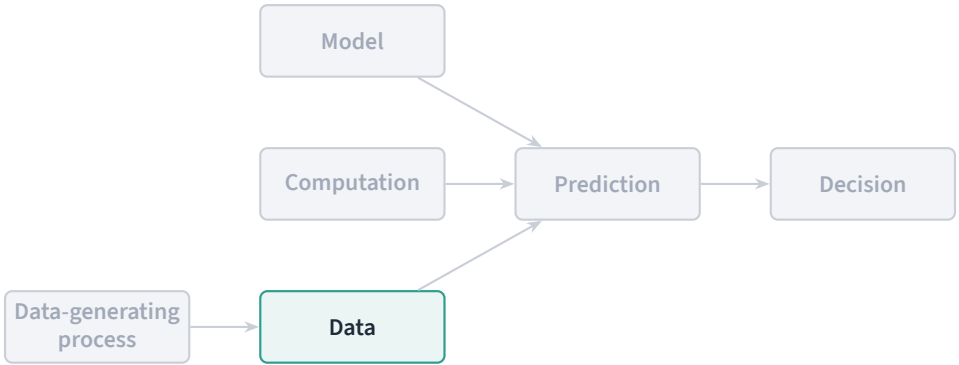
- what are your inputs x, and your outcomes y?
- which inputs do you see, and how often? $p_{\text{true}}(x)$
- how random is the outcome, for a given input? $p_{\text{true}}(y \mid x)$
- could the relationship change over time?

two minutes with your neighbour

Next: the data

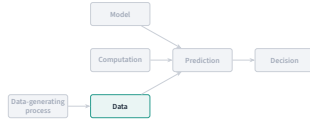


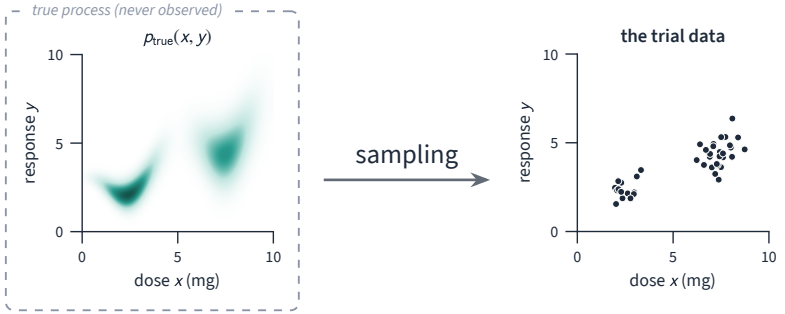
Next: the data



What, how, and how much we measure

Data: samples from the true process

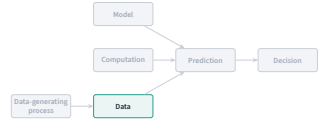


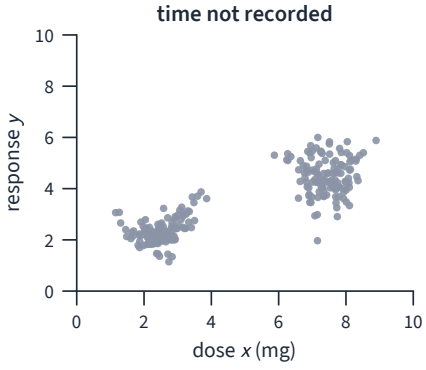


Each patient: a dose $x \sim p_{\text{true}}(x)$, then a response $y \sim p_{\text{true}}(y \mid x)$

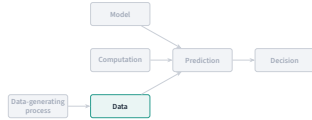
the data $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$; the process itself stays unknown: it is what we try to model from the data

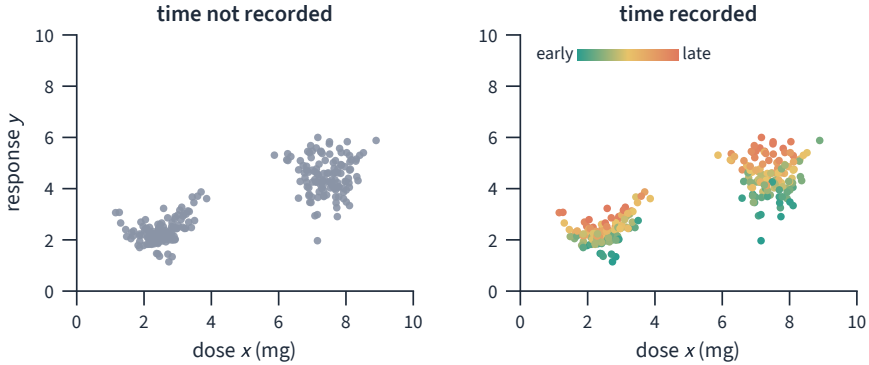
What we measure





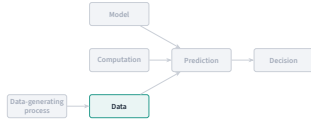
What we measure

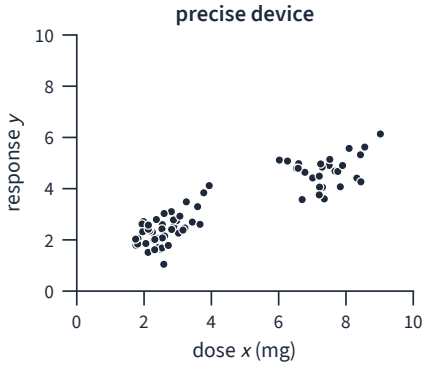




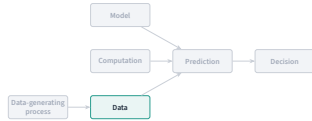
Knowing dose and time reduces variability in the response

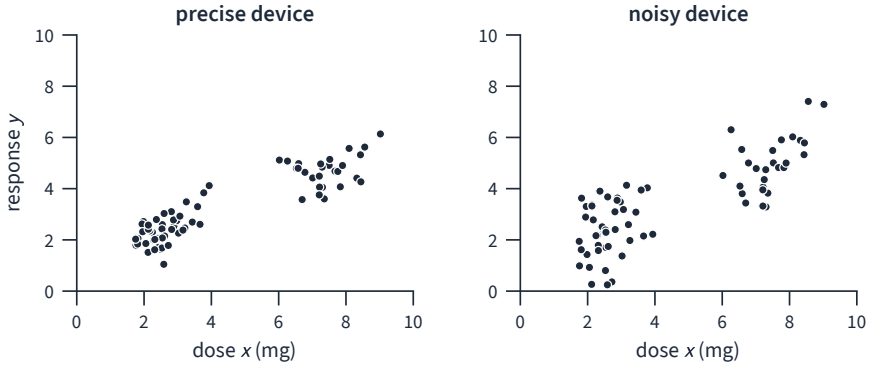
How we measure





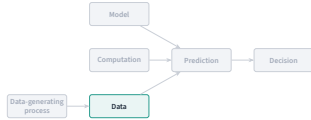
How we measure

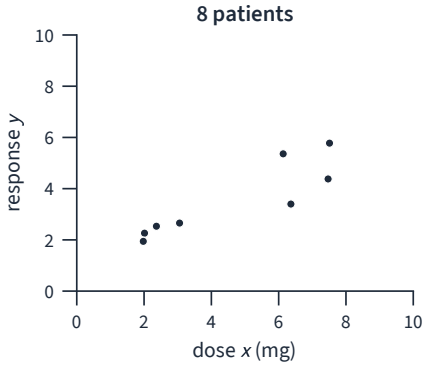




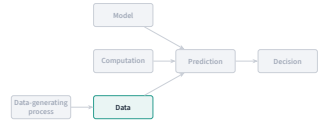
A noisy device increases variability in the measured response

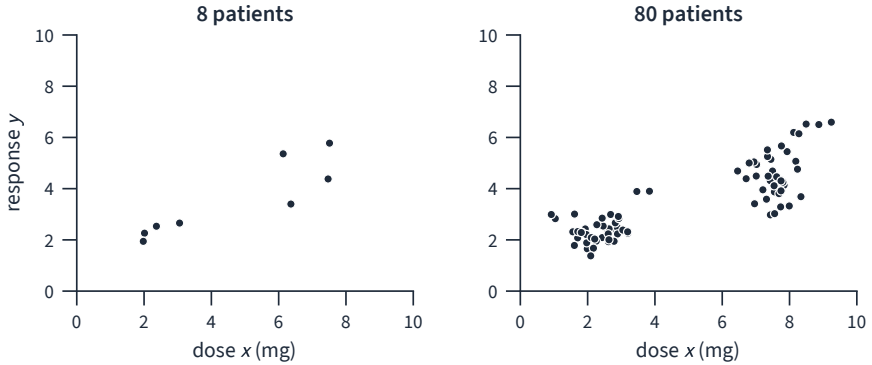
How much: number of doses





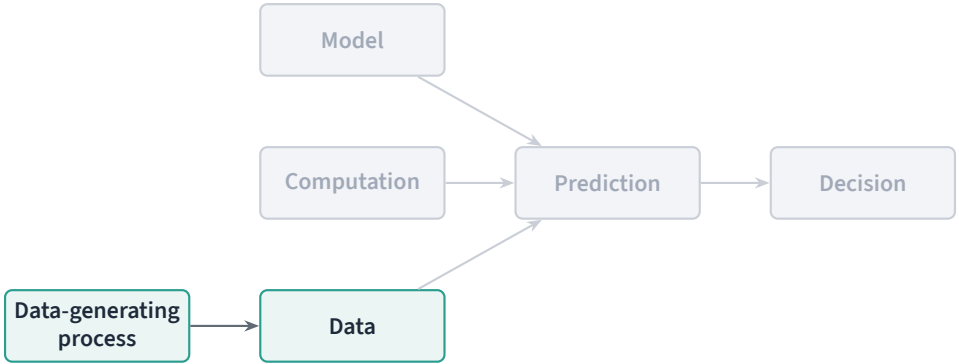
How much: number of doses





More doses: we see which doses occur and how the response changes

Where we are


Model
Computation
Prediction
Decision
Data-generating process
Data

Break